How to set up a free Claude Code CLI with Ollama Local AI Environment.

1. If you are looking for a completely private AI experience, Ollama is an excellent solution. It allows you to download and run various AI models entirely on your local machine. Later in these instructions, we will also ensure the maximum level of security for your data with Ollama.

Note: The following instructions require Windows PowerShell 7 (x64) to be run in Administrator mode.

Step 1: Install Ollama and Download Your Model

First, install the Ollama engine and pull a highly capable, open-weight reasoning model.

PowerShell

Install Ollama via the Windows Package Manager

```
winget install --id=Ollama.Ollama -e
```

To guarantee your data remains secure, you must verify two key settings:

Enable Aeroplane Mode: This ensures that Ollama does not route your queries to the internet or utilise cloud-based models.

Disable Network Exposure: Ensure this setting is turned off so your local AI server refuses incoming API requests from other devices on your network.

Download and run the OpenAI gpt-oss:20b model (requires approx. 14GB of storage and replace 'gpt-oss:20b' with your preferred model if needed)

```
ollama run gpt-oss:20b
```

2. Running Claude Code Locally for Free (Ideal for Computer Science Students). For CS students and developers looking to leverage Claude's exceptional programming capabilities without incurring API or CLI billing costs, you can route Anthropic's Claude Code directly through your private Ollama setup. This also keeps your codebase entirely off the public network.

Step 1: Install and configure the Claude Code

Next: Install and Configure Claude Code and set your environment variables so it communicates with your local machine rather than Anthropic's cloud servers.

Install the Claude Code CLI natively

```
irm https://claude.ai/install.ps1 | iex
```

#Route Anthropic traffic to your local Ollama server (Note: Every time you want to run Claude, you must have to set the environment)

```
$env:ANTHROPIC_AUTH_TOKEN="ollama"
$env:ANTHROPIC_BASE_URL="http://localhost:11434"
```

To switch your model, simply do it from inside Claude.

Launch Claude Code using your local model

```
npx @anthropic-ai/claude-code --model gpt-oss:20b
```

#If you are getting an error because npx is not recognised, you can quickly check if you're set by typing node -v in your terminal. If that doesn't return a version number, you need to install Node.js.

```
npm install -g npm@latest
```